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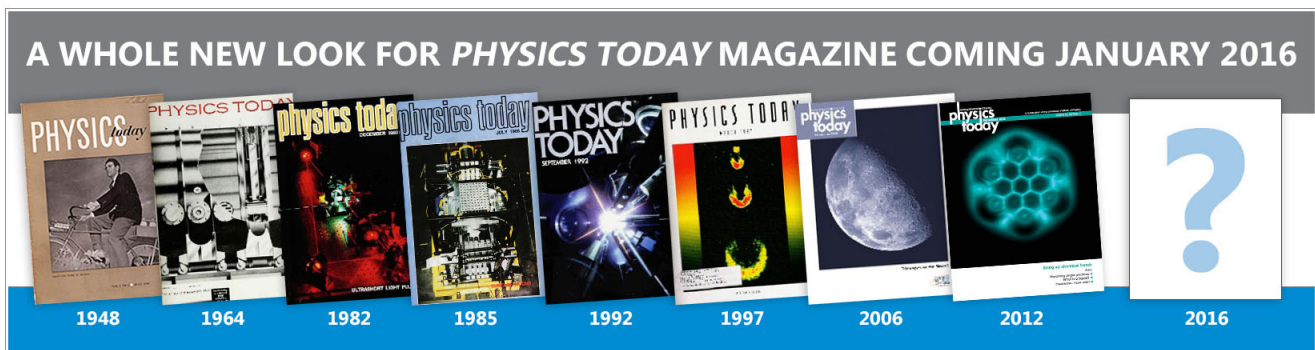
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Incipient air entrainment in a translating axisymmetric plunging laminar jet

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Air entrainment due to a translating axisymmetric laminar water jet plunging into an otherwise quiescent pool of water was studied experimentally. The jet impact diameter, D_j , and velocity, V_j , ranged from 0.33 to 1.21 cm and 104 to 365 cm/s, respectively. For all of these jet flow conditions, the water surface around the jet impact site of the stationary jet is smooth and no air is entrained. When the jet is moving horizontally with velocity V_t , a depression of the free surface forms directly downstream of the jet. In any set of experiments with constant V_j and D_j , as V_t is increased, the depth and streamwise length of this depression increases and a cusp forms at the bottom of its upstream edge. Air entrainment first occurs in the form of discrete small bubbles with diameters of about 0.05 cm that are injected from the cusp when the Froude number ($Fr = V_t / \sqrt{g D_j}$, where g is the acceleration of gravity) exceeds a critical value of about 1.4 for $V_j / V_t > 5$. At higher values of V_t , a separate mode of entrainment occurs where large pockets of air are injected from the cusp into the flow in the form of intermittent bursts. At still higher values of V_t , the high-entrainment-rate condition occurs continuously. High-speed flow visualization movies of the entrainment process show that in all entrainment regimes, the bubbles enter the water when vortices from the jet shear layer pass over the leading-edge cusp of the depression and that these bubbles are initially trapped within the vortices. The boundaries between the three entrainment regimes are determined and presented on a plot of Froude number versus velocity ratio (V_j / V_t). © 2002 American Institute of Physics. [DOI: 10.1063/1.1433493]

I. INTRODUCTION

Air entrainment by a plunging liquid jet into a receiving pool of the same fluid is of particular interest within the geophysical, chemical, and energy sciences due to its importance in aeration of the ocean by plunging breaking waves,¹ its simplicity and efficiency as a gas/liquid reactor,^{2,3} its occurrence behind weirs in open channel flows, and its role in breaking down slugs of gas in two-phase pipe flows.⁴

There is a large body of information on air entrainment in stationary plunging jets and thorough reviews of this work have been given by Chanson,⁵ Wood,⁶ and Bin.⁷ One of the fundamental quantitative results reported in previous studies is the minimum impact velocity required to cause air entrainment, v_c . The minimum impact velocity for circular jets has been studied by Lin and Donnelly,⁸ Sheridan,⁹ Lara,¹⁰ Irvine *et al.*,¹¹ McKeogh and Elsayy,¹² McKeogh and Irvine,¹³ Koga,¹⁴ Bin,¹⁵ Detsch and Sharma,¹⁶ and Wood.⁶ Lin and Donnelly⁸ measured v_c for vertical continuous jets. A range of fluids, nozzle diameters, and standoff distances (L) between the nozzle and the surface of the pool were used. For most of the experimental conditions, the jet was laminar, though a few turbulent cases were also studied. (Even though many of the articles discussed here distinguish between laminar and turbulent jets, the only studies that include measurements of the velocity fluctuation level of the jets are Refs. 11–13. Thus, it seems likely that the authors were assuming

that the jets were laminar or turbulent because the surfaces of the jets were smooth or rough, respectively. In spite of this lack of quantitative information on the velocity fluctuation levels, the laminar and turbulent terminology used by the original authors is preserved in this discussion.) Lin and Donnelly pointed out that the entrainment mechanisms for laminar and turbulent jets appear to be different. In the laminar jet, a sheet of gas was found surrounding the initial submerged region of the jet and bubbles were generated from this sheet. In the turbulent case, entrainment was observed to occur at the pool surface as bumps and valleys on the jet interface passed through the surface of the pool. At incipient conditions for the laminar jet experiments, the data followed a power law relationship between the Weber number ($We = \rho D_j v_c^2 / \sigma$ where ρ is the liquid density, D_j is the jet diameter at the impact site, and σ is the surface tension) and the Reynolds number ($Re = D_j v_c / \nu$ where ν is the kinematic viscosity of the liquid). Sheridan⁹ presented a theory for the surface shape of the liquid pool around the impact site of a laminar vertical water jet. Sheridan theorized that incipient entrainment occurs when the pressure due to the air entrained in the boundary layer around the jet forces the pool surface at the contact point to slope downward toward the jet axis. For jet diameters between 0.2 and 0.4 cm, the theory and Sheridan's experiments with water agree with $v_c \approx 200$ cm/s. At larger diameters (> 0.4 cm), the experimental values of v_c drop while the theoretical value remains unchanged. Lara¹⁰ observed the entrainment for a thin jet (nozzle diameter = 0.058 cm) of water as a function of jet velocity and L . The jet was laminar in all cases; however, the range of parameters

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included cases where the jet broke up into drops before reaching the water surface. Lara observed that for fixed $L/D < 100$ (where D is the nozzle diameter) and increasing jet velocity, entrainment occurs first at low jet velocity due to the impact of droplets created as the jet breaks up before reaching the water surface. As the velocity is increased further, the length of the continuous portion of the jet increases until it reaches the surface of the pool; at this point the entrainment stops. After further increase, entrainment starts for the continuous jet. Thus, for the cases where the jet breaks up before reaching the water surface, the concept of minimum entrainment velocity is not valid. A more detailed study of these effects, including dimensionless correlations of data, has been reported by Bin.¹⁵ Irvine *et al.*,¹¹ McKeogh and Elsayw,¹² and McKeogh and Irvine¹³ reported detailed studies of the inception velocities for circular water jets as a function of the jet diameter and the turbulence intensity. The inception velocity is on the order of 400 cm/s for quiescent laminar flows ($v'/v \leq 0.3\%$, where v' is the rms velocity of the plunging jet). As the turbulence intensity increases, v_c decreases; however, once $v'/v \geq 5\%$, the critical inception velocity maintains a steady value around 80 cm/s. The value of the critical velocity was also found to be relatively insensitive to the diameter of the jet.¹² Koga¹⁴ studied the effect of acute jet impact angles (as measured from the free surface to the jet axis) on v_c as a function of the jet diameter. (Koga did not describe the jets as either laminar or turbulent.) He found that an angular displacement of the jet significantly reduced the critical inception velocity, and proposed that the critical angle, θ_c , has a strong dependence on the Weber number, $We = v_c^2/2\gamma K$ ($\gamma = \sigma/\rho$ and K is the surface curvature). Detsch and Sharma¹⁶ also studied v_c as a function of impact angle, but expanded their study to include several different fluid properties. They observed that the simple relation proposed by Koga was not consistent with their observations of fluids with different surface tensions and viscosities. Although they did not present any analysis of the problem, Detsch and Sharma provided a dimensional correlation which collapsed all of their data.

The case of air entrainment due to a jet plunging into a moving pool is common in both steady and unsteady plunging breaking waves. Unsteady plunging breakers can occur in open bodies of water or laboratory wave tanks. Studies of air entrainment in these waves is generally directed toward the void fraction, bubble size distributions, and radiated noise (see, for example, Refs. 17–19). The data are correlated primarily with characteristics of the wave, such as the wave frequency and “breaking intensity,” rather than the characteristics of the jet itself and the motion of the free surface upon which the jet impacts. Steady breakers occur at the bow of fast moving ships or boats. A similar phenomenon due to a surface-piercing wedge in a shallow water flow has been examined by Waniewski *et al.*²⁰ The water rises up the side of the wedge and peels off into the air, eventually plunging into the flow far from the wedge. Preliminary results point to a relationship between the velocity and roughness length scale of the jet surface to the frequency of bubble production in the underlying flow.

In the present work, the entrainment process for axisym-

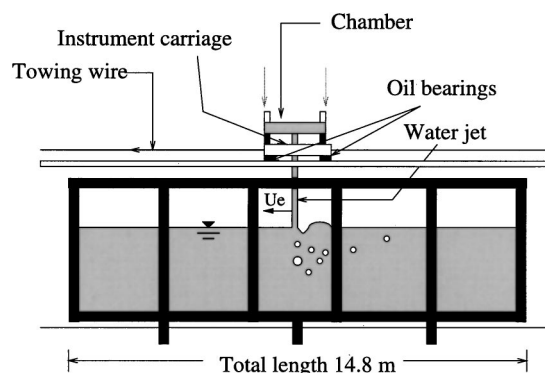


FIG. 1. Schematic side view of the towing tank.

metric steady laminar vertical water jets that are moving horizontally at constant speed over a quiescent pool of water is examined experimentally. (This case is not intended as a model of a breaking wave where the jet is two dimensional and air entrainment starts with an initial jet impact. Also, results for a two-dimensional steady jet are expected to be quite different from those presented herein.) In all the jet conditions of the current study, no air is entrained when the jet is stationary. When the jet is moving horizontally, four entrainment regimes (no entrainment, incipient entrainment, intermittent entrainment, and continuous entrainment) are observed. The three boundaries between these regimes are determined in terms of the horizontal velocity, the jet impact diameter, and the jet vertical impact velocity. Flow visualization with high-speed movies is used to elucidate some aspects of the entrainment process.

The remainder of this paper is divided into three sections. The experimental apparatus and procedures are described in Sec. II. This is followed in Sec. III by a presentation and discussion of the experimental results. Finally, the conclusions are given in Sec. IV.

II. EXPERIMENTAL APPARATUS AND TECHNIQUES

The experiments were performed in a tank that is 1480 cm long, 122 cm wide, and 100 cm deep (see Figs. 1 and 2). The sidewalls of the tank are made of glass to allow for flow visualization and optical measurements. An instrument car-

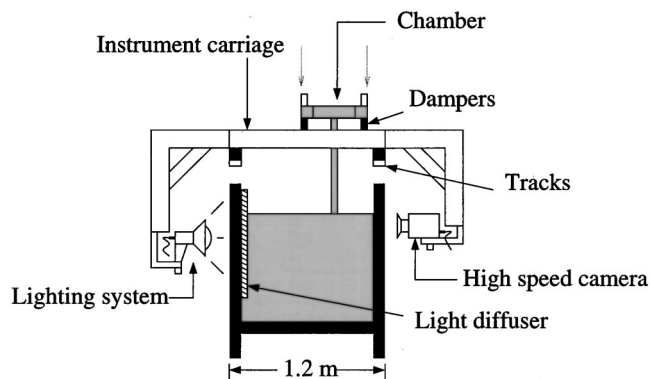


FIG. 2. Schematic end view of the towing tank.

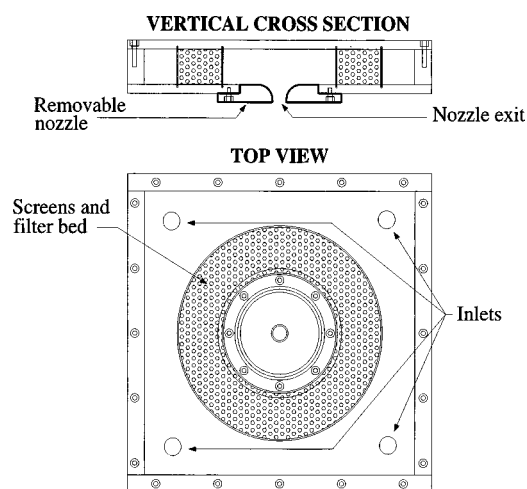


FIG. 3. Jet nozzle assembly.

riage rides horizontally along the length of the tank on two tracks, one above each of the long sidewalls. The carriage is supported above the tracks via four hydrostatic oil bearings, thereby greatly reducing vibration and friction levels of the carriage. The carriage is driven by two towing wires which are powered by a servo motor. A computer controls the starting position, acceleration, and velocity of the carriage. Precise towing speeds are obtained by means of a computer-based feedback control system. The accuracy of the input towing speed was verified with an independent measurement obtained from two timing switches located 100.0 cm apart and connected to a timer with nanosecond accuracy. The towing speed was found to be accurate to within $\pm 1\%$.

The jet was created from a nozzle located on the bottom face of an acrylic chamber (see Fig. 3). The chamber was mounted on top of the carriage such that the jet impinged vertically down onto the free surface of the water. Three interchangeable nozzles with outlet diameters of 1.27, 0.72, 0.41 cm were used. Each nozzle was machined from a 1.27-cm thick Plexiglas™ plate. In order to minimize the thickness of the jet boundary layer at the nozzle exit, the profile of the inlet of each nozzle was rounded with a circular shape and the length of the constant-diameter outlet section was relatively short (about 0.6–1 cm depending on the nozzle, see Fig. 3). The chamber plenum was simultaneously fed by four water inlets (one at each corner of the top face), to ensure a uniform supply to the exit. A 5-cm-thick annulus surrounding the nozzle was bounded by two metallic screens and filled with spherical glass beads (approximately 0.3 cm in diameter) in order to provide a uniform flow.

Water was supplied to the chamber by a centrifugal pump mounted on the side of the carriage. To eliminate jet instabilities caused by pump-induced carriage vibrations, the pump and the jet nozzle chamber were mounted on dampers. The intake pipe for the pump extended vertically into the water from the top of the carriage and was located about 100 cm downstream of the jet chamber. To measure the flow rate of the system, a rotameter (Omega Engineering Inc. model FL1503A) with a maximum flow rate of 160 cm³/s was placed on the outlet side of the pump. The rotameter was

calibrated by measuring the volume of water pumped over a measured time interval. The accuracy of the flow rates reported herein is estimated to be $\pm 3\%$. The above system with the nozzle mounted at a range of heights above the tank produced jets with impact diameters, D_j , ranging from 0.33 to 1.21 cm, and impact velocities, V_j , ranging from 104 to 365 cm/s, at the free surface.

In order to characterize the jet, the centerline mean velocity (v) and root-mean-square velocity fluctuation (v') were measured using a single-component laser-Doppler velocimeter (LDV). The LDV was set up so that the plane formed by the two laser beams was vertical and passed through the jet centerline. In order to minimize the measurement volume, the collection optics were placed perpendicular to the transmitter optics. It is estimated that the cylindrical measurement volume (the intersection of the two beams and the focal point of the collection optics) was about 150 μm in diameter with a length of about 150 μm . In order to accommodate the LDV traverse system, the jet nozzle chamber was taken off the carriage and placed in a separate pump loop facility. The nozzle was placed at a height above the receiving pool greater than any of the heights used in the entrainment experiments. In order to provide scatterers for the LDV measurements, the water supply was seeded with silver-coated hollow glass spheres with average diameters of 12 μm . For each of the jet conditions corresponding to the 21 entrainment measurements reported in the following, velocities were measured at positions 3 cm below the nozzle and at the equivalent location of the water surface in the entrainment experiments. It should be emphasized that several phenomena can contribute to the measured velocity fluctuation level. These phenomena include decaying turbulence from the nozzle and plenum, oscillations due to capillary instability, transient jet oscillations caused by various mechanical factors, movement of the measurement volume off the jet centerline by jet surface oscillations and a slow migration of the average jet velocity during the measurement period. In this study, the total range of v'/v over all 42 measurements was from 0.20% to 0.35%. In view of this low fluctuation level and the fact that at least some portion of each measurement was due to factors other than jet turbulence, the jet is called laminar in this paper.

Observations of the water jet and air entrainment process were carried out with a high-speed digital camera (Vision Research, Phantom V3.0) which was mounted on the side of the carriage and viewed the flow from above or below the free surface through the glass sidewall of the tank. For several experimental conditions, solid titanium dioxide particles with an average diameter of 1.24 μm were injected into the pump inlet so that the jet became visible when viewed from below the water surface. Observations of the entrainment boundaries as described in the following section were carried out with frame rates of 500 Hz while movies of the jet seeded with the titanium dioxide particles were taken at 1635 Hz. The flow was illuminated by a continuous light source that was mounted on the carriage opposite to the camera (see Fig. 2). A light diffuser screen was attached to the glass wall of the tank nearest to the light in order to make the illumination more uniform. In order to obtain the illustrative pho-

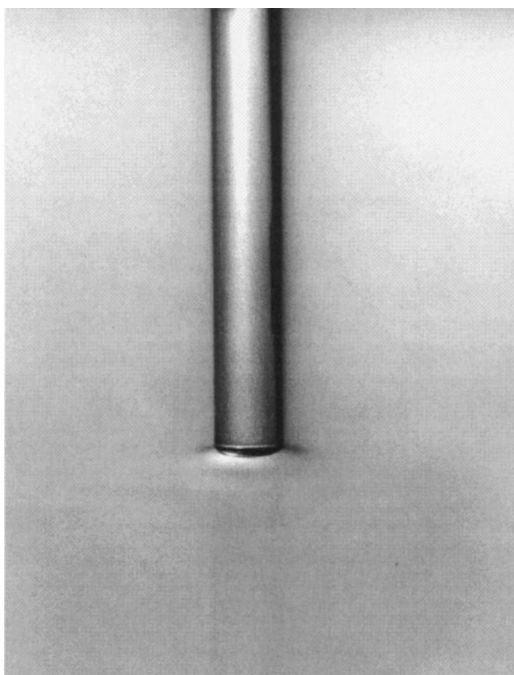


FIG. 4. Above-water view of the jet impact site. $V_j = 303$ cm/s, $D_j = 0.6$ cm, $V_t = 0.0$ cm/s.

tographs shown in Figs. 4 and 5, the digital camera was replaced with a 35 mm film camera (Nikon F3) and the continuous light source was replaced with a strobe with a duration of 3.3×10^{-5} s. The images were recorded on Kodak Tmax-100 black-and-white negative film.

A filtration system was used to maintain water quality in the tank and remove the titanium dioxide particles following the jet visualization experiments. In this system, the water was removed from the surface by two gravity-fed skimmers and sent to a diatomaceous earth filter. The water was then recirculated back into the tank at the end opposite the skimmers. The surface tension in the tank was monitored with a Wilhelmy plate in combination with a Langmuir trough as described in Duncan *et al.*²¹ The Langmuir trough was used to compress the water surface, thus increasing the surface concentration of surfactants around the Wilhelmy plate. For all experiments reported herein, the ambient surface tension was close to 73×10^{-3} N/m and remained at this value while the surface was compressed by a factor of 4.

III. RESULTS AND DISCUSSION

A. General observations

A photograph of the stationary jet ($U_t = 0$) impinging on the water surface is shown in Fig. 4. The jet impact diameter and velocity ($D_j = 0.6$ cm, $V_j = 303$ cm/s, respectively) are typical of the experimental conditions used herein. As can be seen from the photograph, the jet surface appears very smooth and a thin region of the water surface around the jet impact site rises slightly around the jet. This meniscus effect has been noted by a number of authors including Sheridan⁹ and Lienhard²² and is discussed by Bin.^{15,7} When the jet velocity is low, stationary capillary waves can also be seen on the jet surface above the meniscus. Theory and experi-

ments exploring these waves have been reported by Lienhard.²² No air is entrained and no measurable stationary capillary waves appear when the jets are stationary for any of the jet velocity and diameter combinations used in this study. The fact that no entrainment was found for speeds up to 365 cm/s in the present experiments where the maximum jet velocity fluctuation level was $\leq 0.35\%$ is consistent with Refs. 11–13 where incipient entrainment occurred at a jet velocity of 360 cm/s with a jet velocity fluctuation level of 0.3%.

Figures 5(a)–5(f) contain photographs of the jet impact taken with above-surface and below-surface views for $D_j = 0.6$ cm, $V_j = 303$ cm/s and three separate translational speeds, V_t . A schematic of the flow field, including the nomenclature used in the following discussion, is presented in Fig. 6. The above-surface views in Figs. 5(a) ($V_t = 31.75$ cm/s), 5(c) ($V_t = 38.10$ cm/s), and 5(e) ($V_t = 44.45$ cm/s) show the smooth jet impacting on the water. A capillary wave pattern appears upstream of the jet and the length of the capillary waves decreases as V_t increases. A depression in the free surface can be seen directly downstream of the jet, which forms the upstream end of a relatively long-wavelength gravity-capillary wave train. The streamwise length of the depression and the centerline wavelength of the gravity-capillary wave train increase as V_t is increased. The small-scale roughness of the water surface behind the jet is an indication of a subsurface turbulent wake.

Underwater views corresponding to the photographs in Figs. 5(a), 5(c), and 5(e) are given in Figs. 5(b), 5(d), and 5(f), respectively. The jet is not visible in the photographs since it was not marked with particles in this particular sequence of experimental runs. However, high-speed movies with particles in the jet show that the steep upstream edge of the depression forms on the downstream edge of the jet (see Sec. III C and Fig. 12). For the lowest speed, Fig. 5(b), the surface depression is relatively shallow and of limited streamwise length; no air is entrained. At the intermediate speed, Fig. 5(d), the depression is longer and extends deeper into the water. There are a number of undulations on the upstream, nearly vertical edge of the depression, and a cusp has formed at the bottom of this leading edge. In the high-speed movies, it is found that this cusp extends downward periodically and, occasionally, a small bubble is pinched off. The bubble is then entrained into the turbulent jet flow where it travels deeper into the water and translates downstream. One such bubble with diameter ≈ 0.05 cm can be seen in Fig. 7 which is a magnified view of the cusp region of Fig. 5(d). In the following, this condition is called the incipient entrainment regime. For the highest translation speed, Fig. 5(f), the surface depression is deeper and longer. A large number of bubbles are being entrained at the time of the photograph. These bubbles enter the turbulent jet flow where they are broken into smaller bubbles by turbulence and shear flow processes (see, e.g., Refs. 23–28). For this translation speed, the air entrainment process starts and stops periodically and this condition is called the intermittent entrainment regime. At still higher translational velocities, the air entrainment occurs without interruption and this condition is called the continuous entrainment regime.

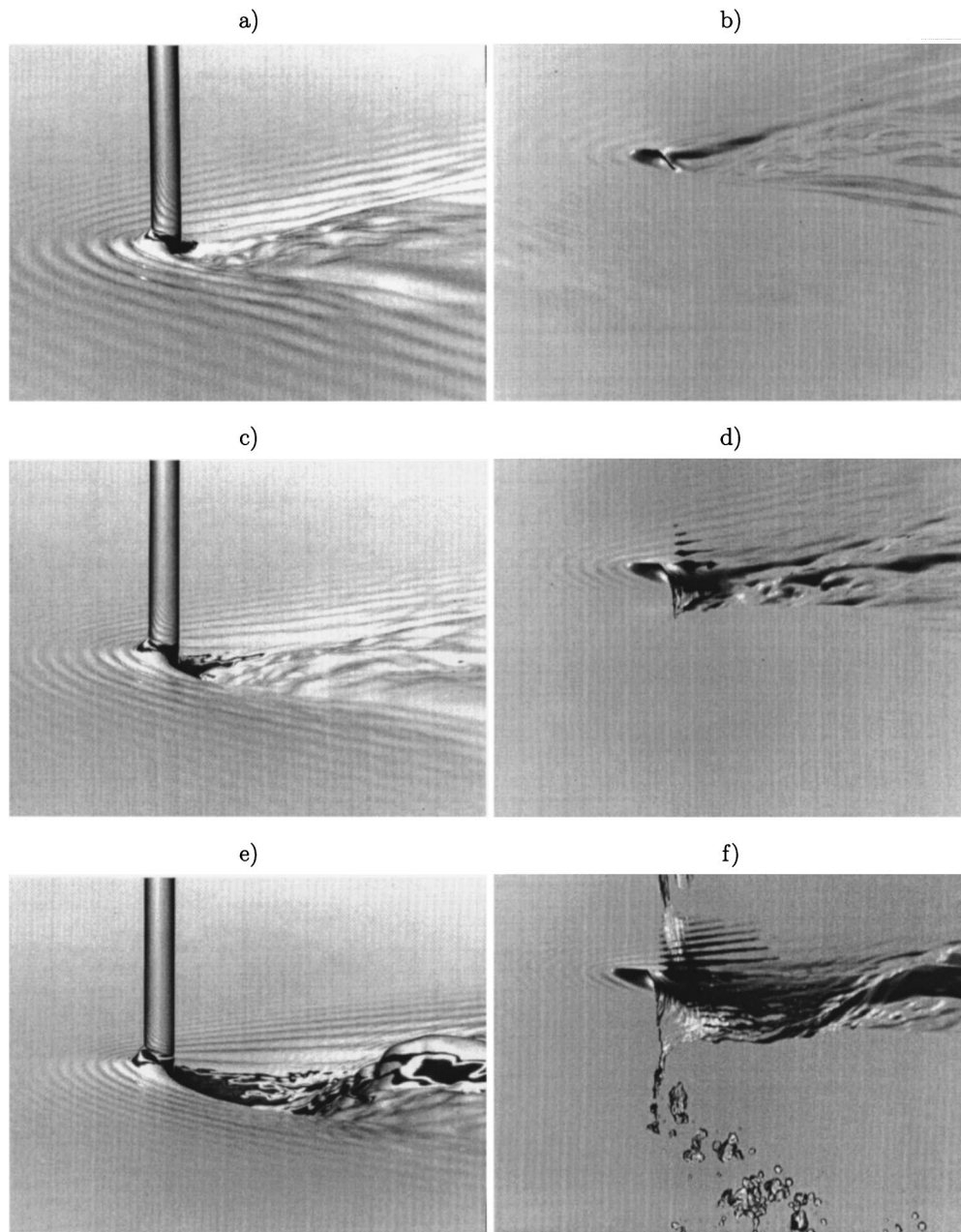


FIG. 5. Above-surface (left column) and below-surface (right column) views of the jet impact site for three translational velocities. $V_j = 303$ cm/s, $D_j = 0.6$ cm. (a) and (b): $V_t = 31.8$ cm/s, (c) and (d) 38.1 cm/s, (e) and (f): $V_t = 44.5$ cm/s.

B. Entrainment boundaries

A set of experiments was performed to locate the three boundaries between the four entrainment regimes (no entrainment, incipient entrainment, intermittent entrainment, and continuous entrainment) in terms of D_j , V_j , and V_t . The other parameters of the entrainment process, namely gravity and the properties of the liquid (water), were held constant. To find the boundaries, 21 different combinations of D_j and V_j were selected, see Table I. For each combination, a set of underwater high-speed movies of the jet impact were taken for different values of V_t . The values of V_t were selected iteratively until the critical velocity for each boundary was found. The boundary between no entrainment and incipient entrainment was defined as the smallest V_t for

which at least one small bubble was entrained during a 2 s period (1000 frames of the high-speed movie). This is a sensitive measurement and many repeated runs were taken at conditions close to the entrainment boundary in order to determine the boundary accurately. The boundary between incipient and intermittent entrainment was found by locating the smallest V_t that produced intermittent groups of entrained bubbles. The boundary between intermittent and continuous entrainment was taken as the minimum value of V_t that produced a high bubble entrainment rate without noticeable interruption.

In the most general case of air entrainment by liquid flow, a large number of dimensional parameters can influence the entrainment process. Reviews of these generalized

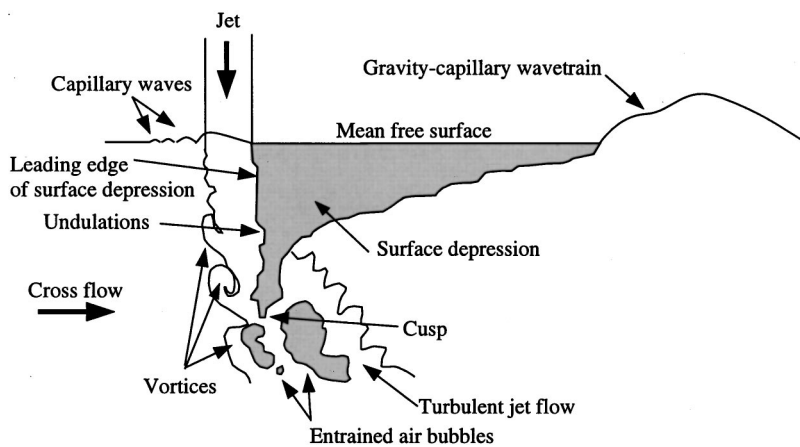


FIG. 6. Nomenclature for the entraining jet flow shown in the reference frame moving with the jet. The cavity, bubble and jet profiles were taken from the last frame of the image sequence shown in Fig. 12.

considerations are given by Chanson⁵ and Wood.⁶ For the specific problem of a laminar, translating jet, the entrainment conditions would depend on a reduced set of parameters determined by the intrinsic properties of the two fluids (the liquid density, ρ_l , the gas density, ρ_g , the liquid viscosity, μ_l , the gas viscosity, μ_g , and the surface tension, σ) as well as those imposed by the flow boundary and initial conditions (gravitational constant, g , the diameter of the jet, D_j , the impact velocity, V_j , and the translation velocity, V_t). This provides nine dimensional parameters, which can be reduced to six independent nondimensional parameters. Using the liquid density, the translation velocity, and the jet diameter as the representative repeating variables, a Froude number ($Fr = V_t/\sqrt{gD_j}$), a Weber number ($We = \rho_l D_j V_t^2/\sigma$), a liquid cross-flow Reynolds number ($Re_l = \rho_l D_j V_t/\mu_l$), a mixed Reynolds number ($Re_m = \rho_l D_j V_t/\mu_g$), a velocity ratio ($R = V_j/V_t$), and a density ratio, ($\gamma = \rho_g/\rho_l$) are produced from the procedure. Perhaps a more physically relevant dimensionless number than the mixed Reynolds number Re_m is the viscosity ratio μ_l/μ_g obtained by dividing Re_m by Re_l .

From the perspective of the experiments, only three dimensional parameters could be independently controlled: V_t , V_j , and D_j . Since both working fluids (air and water) were kept constant, both the density ratio and viscosity ratio were fixed. The functional relationships between the remaining dimensionless ratios were examined, in addition to numerous combinations of these terms, with the best correlations by far being produced by plotting Fr vs R as shown in Fig. 8. The correlation in the figure implies that the inception process is dominated by inertial and gravitational effects and this idea is supported by the relatively large Weber numbers ($We = V_j^2 D_j/\sigma$ varying from 91.9 to 932.5), Reynolds numbers ($Re_j = V_j D_j/\nu$ varying from 5370 to 21820), and Bond numbers ($B = \rho_l g D_j^2/\sigma$ varying from 1.46 to 19.67) in the data. It is presumed that if it were possible to do the experiments with a different range of parameters including the use of several fluids with a wide range of surface tensions and viscosities, the dependence of the results on other dimensionless numbers would become apparent. The boundary between no entrainment and incipient entrainment (solid line) is nearly horizontal to the right of $V_j/V_t = 5.0$ and dips slightly at the lowest velocity ratios. The average Froude number of this boundary is 1.4 with a standard deviation of ± 0.1 . The

incipient/intermittent (long dash) and the intermittent/continuous (short dash) entrainment boundaries are nearly vertical at $V_j/V_t = 6$ and the slope of these boundaries decreases for $V_j/V_t < 5$ where all three boundaries are close together and parallel. It is speculated that the coincidence of the entrainment boundaries at low jet velocity ratios is associated with the tendency of the underwater trajectory of the jet to bend further downstream into the surface depression tip in those cases. This tendency was verified in the high-speed movies taken with the jet marked with particles. It is important to note that for velocity ratios higher than about 8, very high Froude numbers would be required to produce intermittent or continuous air entrainment.

The present translating jet experiments are quite different from the experiments of Koga¹⁴ and Detsch and Sharma¹⁶ where the effect of the jet inclination angle on the incipient entrainment conditions for a stationary jet was studied. In these previous experiments, the jet nozzle and the bulk of the receiving pool were stationary so the axis of the underwater jet plume (without the buoyancy effect of the bubbles) was most likely along the direction of jet impact. The entrainment of air occurred at the point where the upper surface of the jet entered the water. The critical angles for air entrainment (as measured from the horizontal to the jet axis) were between 20° and 60° for nozzle diameters ranging from 0.10 to 0.46 cm. In the present experiments, the air is en-

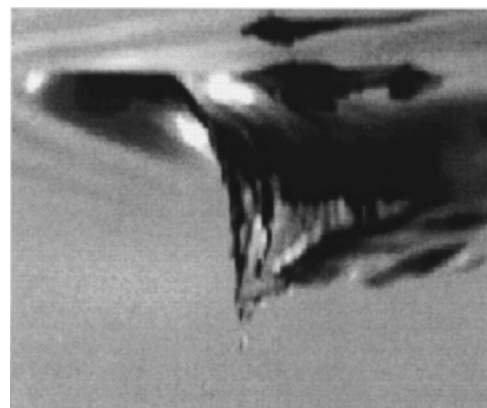


FIG. 7. Magnified view of the cusp region of Fig. 5(d) showing a small bubble.

TABLE I. Experimental conditions and entrainment boundaries.

Experiment condition	Nozzle diameter D_n (cm)	Jet height h (cm)	Jet flow rate Q (cm ³ /s)	Impact diameter D_j (cm)	Impact velocity V_j (cm/s)	Translation velocity boundary		
						Incipient V_{ic} (cm/s)	Intermittent V_{it} (cm/s)	Continuous V_c (cm/s)
1	0.72	18.8	32.1	0.45	204	30.5	34.3	38.1
2	0.72	18.8	60.1	0.57	237	33.7	38.1	41.9
3	0.72	18.8	98.2	0.65	298	35.6	43.2	47.0
4	1.27	18.8	60.7	0.63	192	35.6	38.1	40.6
5	1.27	18.8	107.4	0.83	201	40.0	41.9	43.2
6	1.27	18.8	81.6	0.71	207	36.8	39.4	41.3
7	0.41	19.7	31.6	0.36	315	27.3	45.7	50.8
8	0.41	34.0	31.6	0.33	365	25.4	45.7	52.1
9	0.72	33.4	32.1	0.39	272	25.4	40.6	44.5
10	0.72	33.4	60.2	0.52	283	28.6	41.9	45.7
11	1.27	33.4	60.2	0.54	260	33.0	41.9	47.0
12	1.27	33.3	31.8	0.40	256	28.6	40.6	44.5
13	1.27	33.3	81.8	0.63	264	34.3	40.6	41.9
14	1.27	33.3	108.2	0.72	269	36.8	43.2	44.5
15	1.27	33.3	159.3	0.85	283	38.7	45.7	47.0
16	1.27	2.8	248.9	1.21	216	43.2	45.7	48.3
17	1.27	2.8	121.6	1.15	118	37.5	38.7	40.6
18	0.72	2.7	31.4	0.62	104	30.5	33.0	...
19	0.72	2.7	60.4	0.67	172	34.3	36.2	37.5
20	0.72	2.7	107.7	0.70	282	38.1	43.2	45.7
21	0.41	3.6	17.6	0.37	161	29.8	29.8	31.1

trained at the bottom edge of the surface depression which is created in part by the free stream motion as seen from the reference frame of the jet. The entrainment point is on the opposite side of the jet from that in the steady inclined jet experiments. At the entrainment boundaries, the angle between the jet velocity vector [measured in the reference frame of the water at infinite depth, $\alpha = \tan^{-1}(V_j/V_t)$] and the horizontal ranged from 71° to 86° .

C. Entrainment visualizations

To achieve a better understanding of the entrainment boundary data, a set of visualization experiments was performed with the jet marked by particles. Four of the 21 above-mentioned D_j - V_j combinations were selected and for each combination four movies were taken at conditions corresponding to pre-, incipient, intermittent, and continuous entrainment. A number of general observations were made from the movies. First, vortices forming on the shear layer surrounding the entraining jet can be seen on the upstream side of the jet just below the free surface under all conditions (see Fig. 6). The vortices gain strength with distance from the free surface and travel along the jet profile as it penetrates vertically and downstream into the water. These vortices are typical of those observed in jets in cross flows with the free surface replaced by a solid wall, see, e.g., Fric and Roshko.²⁹ Second, the above-mentioned small undulations on the leading vertical edge of the surface depression behind the jet also travel downward. Third, as was mentioned in Sec. III A, the depth of the cusp of the depression increases with increasing V_t . Finally, near incipient entrainment conditions the cusp oscillates vertically.

In order to analyze the relationship between the vortices and the undulations of the cusp, several image sequences

were processed to determine the temporal variations of the vertical positions of the vortex centers, the peaks in the undulations on the leading edge of the surface depression, and the cusp tip. These positions were determined by displaying each image on a computer screen and locating each position by eye with the cursor. The undulations on the upstream edge of the depression were frequently difficult to identify due to their small amplitude. In Fig. 9, the vertical positions of the vortices, surface undulations, and cusp are plotted versus time for an incipient entrainment case. The vertical position of the vortices and undulations follow straight lines with approximately the same slope indicating that they have the same vertical velocity. The depth of the cusp tip oscillates

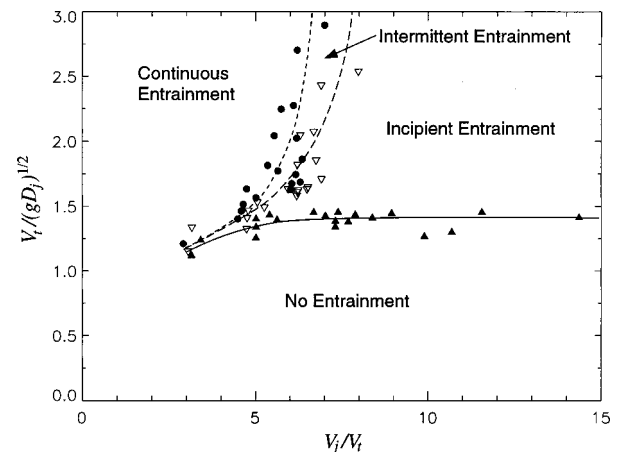


FIG. 8. Froude number ($V_t / \sqrt{g D_j}$) vs velocity ratio (V_j / V_t) for the three entrainment boundaries: $\blacktriangle$ and (—), no/incipient entrainment; ∇ and (---), incipient/intermittent entrainment; $\bullet$ and (---), intermittent/continuous entrainment.

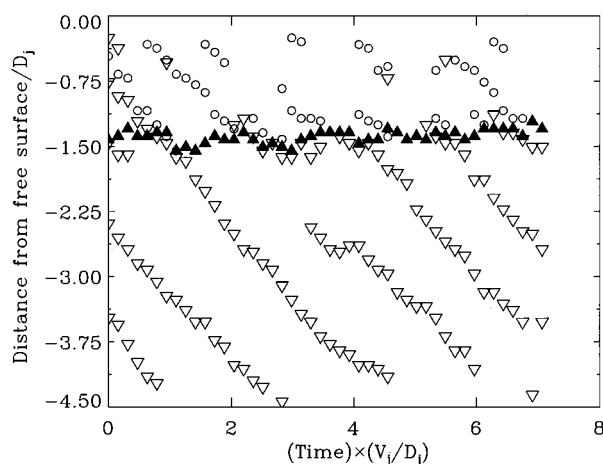


FIG. 9. Temporal evolution of the depth of vortices, undulations on the leading edge of the cavity, and the cusp of the cavity for an incipient entrainment case. $D_j = 0.67$ cm, $V_j = 172$ cm/s. ∇ — vortex centers, $\circ$ — undulations on cavity leading edge, $\blacktriangle$ — cusp.

slightly around a constant value; the period of this oscillation is the same as the period associated with the leading edge undulations and vortices passing a fixed depth. The cusp moves downward with the passage of each vortex. A similar plot for a different time period and the same incipient entrainment condition is given in Fig. 10. In this second plot, a single bubble is entrained at a time of about $3D_j/V_j$ and its vertical position is tracked versus time. As can be seen from Fig. 10, the bubble is entrained during one of the downward excursions of the cusp. This behavior is typical of all the bubble entrainment events at incipient entrainment conditions. The trajectory of the bubble is nearly parallel to that of the vortex that passed over the cusp at the time of the bubble entrainment. In agreement with this data, the high-speed movie shows that the bubble is entrapped within the vortex.

The above-mentioned findings point to the important roles of the cusp depth and the jet vortices in the incipient entrainment process. The trajectory data indicate that the un-

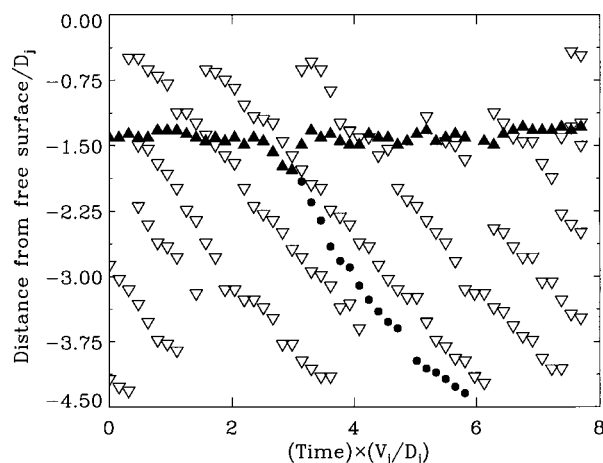


FIG. 10. Temporal evolution of the depth of vortices, the cusp of the cavity, and a single small bubble for an incipient entrainment case. The undulations on the leading edge of the cavity were too small to identify. $D_j = 0.67$ cm, $V_j = 172$ cm/s. ∇ — vortex centers, $\blacktriangle$ — cusp, $\bullet$ — bubble.

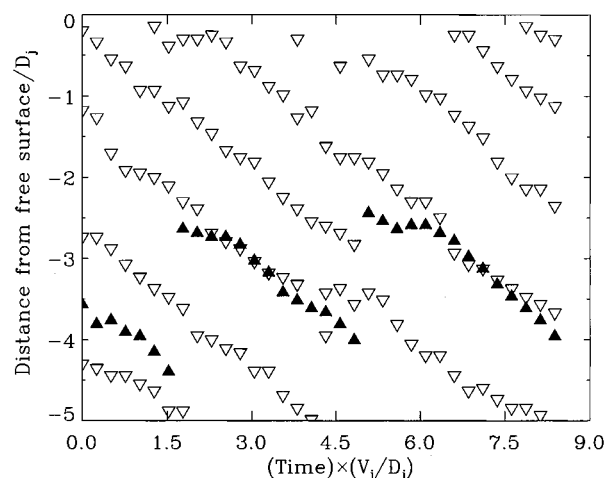


FIG. 11. Temporal evolution of the depth of the vortices, and the cusp of the cavity for an intermittent entrainment case. $D_j = 0.57$ cm, $V_j = 237$ cm/s. ∇ — vortex centers, $\blacktriangle$ — cusp.

dulations on the leading edge of the surface depression and the oscillations of the cusp are manifestations of the jet vortices which wrap around the jet and probably end on the leading edge of the interface. As V_t is increased, the depth of the surface depression increases and pushes the cusp into regions of the jet where the vortices are more developed. The lower pressures in these developed vortices increase the amplitude of the oscillations of the cusp. At incipient entrainment, the cusp depth has reached the point where the pressures in the vortices are low enough to entrain small bubbles into the water.

A plot of the depths of the cusp and the vortices versus time for an intermittent entrainment condition is shown in Fig. 11. The undulations on the leading edge of the interface were not plotted because they were difficult to determine in the movie. The data were taken during a period of entrainment. As can be seen from Fig. 11, the cusp undergoes vertical excursions of about $1.5D_j$. The motion consists of a period of nearly constant downward speed followed by a sudden upward jump. The upward jump is the result of a pinching off of a large bubble rather than a large sudden vertical motion of the cusp. The vortex trajectories have the same velocity and phase as the cusp trajectories, indicating that the cusp motion is likely caused by the vortices. A vortex pairing occurs at a time of about $5D_j/V_j$ when a bubble is pinched off; however, the coincidence of these two events occurs only occasionally. A sequence of five frames from the corresponding high-speed underwater movie is shown in Fig. 12. The schematic drawing in Fig. 6 is taken from the last frame of this sequence. In the first frame of the sequence, a bubble is forming at the cusp. This bubble is at the same height as one of the vortices which can be seen on the upstream side of the jet. In the next two frames, a finger forms on the upstream side of the bubble as it grows and moves downward along with the vortex. In the last two frames, the bubble pinches off from the cusp and the finger pinches off from the main bubble and moves up into the upstream side of the vortex. Though the images show only a silhouette of a three-dimensional phenomenon, it appears that the finger is

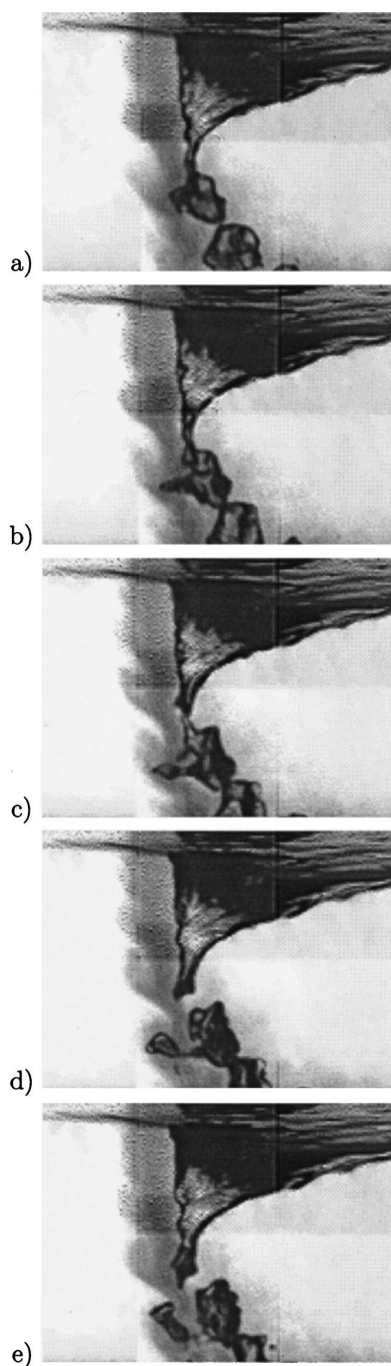


FIG. 12. Sequence of images of a bubble forming inside a jet vortex. The jet flow is seeded with small particles. The image sequence was taken with a high-speed digital camera. Time between frames = 1.2 ms. $D_j = 0.57$ cm, $V_j = 237$ cm/s, $V_t = 38.1$ cm/s.

moving around the periphery of the jet in the center of the ring vortex.

In order to further assess the influence of the downward motion of the jet and the jet vortices on the entrainment process, an underwater view of the free-surface interface produced by a solid, surface-piercing rod is shown in Fig. 13. The towing speed and rod diameter are identical to the speed and impact diameter for the intermittent entrainment condition shown in Fig. 5(f), $V_t = 44.5$ cm/s and $D_j = 0.6$ cm. A comparison of the two photographs shows that in the case of

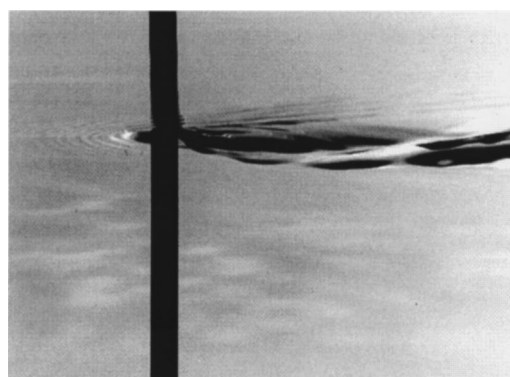


FIG. 13. Below-water view of a surface-piercing solid rod. The upper part of the image of the rod is a reflection from the water surface. The rod diameter and the towing speed are the same as the jet impact diameter and velocity in Fig. 5(f), $D_j = 0.6$ cm, $V_t = 44.5$ cm/s.

the rod, the depth of the surface depression is much smaller, no cusp is formed at the leading edge of the interface, and there is no air entrainment. This comparison indicates the important role played by the vertical momentum and large-scale flow structures of the jet in dramatically modifying both the mean and fluctuating interface dynamics.

IV. CONCLUSIONS

Air entrainment due to a translating, axisymmetric, vertical, plunging, laminar jet was explored. For conditions where the jet does not entrain air when stationary, air entrainment can be induced by increasing the horizontal speed (V_t) to only a small fraction of the jet velocity (V_j). The measurements show that the entrainment is produced by a combination of the tendency of any horizontally moving surface-piercing object to create a downstream surface depression and the influence of the downward motion of the jet as it becomes turbulent after entering the water pool. In particular, the depth of the surface depression and the formation of a cusp at the bottom of the depression's leading edge is directly linked to the jet underwater flow field. Pockets of air are entrained into the flow as vortices from the shear layer around the jet pass over the cusp. These pockets tend to enter the vortex centers and travel around the jet. Three different entrainment regimes were found: incipient entrainment where single small bubbles (diameter ≈ 0.05 cm) are occasionally entrained; intermittent entrainment where the flow oscillates between states where large numbers of larger bubbles are entrained and states of incipient entrainment; and continuous entrainment where the higher entrainment conditions occur without interruption. The boundaries between these entrainment regions are best correlated on a plane of $Fr = V_t / \sqrt{g D_j}$ vs V_j / V_t . The boundary between incipient entrainment and no entrainment occurs at nearly constant $Fr = 1.4$ for $V_j / V_t > 5$ and this critical value decreases to about 1.1 at $V_j / V_t \approx 3$. For jet velocity ratios less than about 6, the boundaries between the entrainment regions are nearly coincident while at higher jet velocity ratios the region of incipient entrainment becomes quite large.

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